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# Abelian Squares are Avoidable on 4 Letters

Veikko Keränen

Rovaniemi Institute of Technology
Jokiväylä 11, SF-96300 Rovaniemi, Finland

Abstract. *An* abelian square *is a non-empty word of the form* $P_1P_2$, *where* $P_2$ *is a permutation of* $P_1$, *that is, the number of occurrences of each letter is the same in* $P_1$ *and* $P_2$. *A word is called* abelian 2-repetition free, *or in short* a-2-free, *if it does not contain any abelian square as a subword. Let* $\Sigma$ *be the four letter alphabet* $\{a,b,c,d\}$. *We give an example of a uniformly growing endo-morphism* $g: \Sigma^* \to \Sigma^*$; *with* $|g(a)| = 85$; *such that the iteration of* $g$ *yields an a-2-free* $\omega$*-word. In fact, it is proved that the morphism* $g$ *itself is* a-2-free, *that is,* $g(w)$ *is a-2-free for every a-2-free word* $w$ *in* $\Sigma^*$. *When proving that this property holds for* $g$, *one is in practice obliged to use a computer for certain tests. We performed these tests many times by using different computing environments and methods. It is explained how rechecking can be efficiently accomplished.*

*The morphism* $g$ *is of the form* $g(\sigma(x)) = \sigma(g(x))$ *for all* $x$ *in* $\Sigma$, *where* $\sigma$ *performs a cyclic permutation of letters in* $\Sigma$. *As regards morphisms of this form, one may check by using a computer/computers that, for our example morphism* $g$, *the image word* $g(a)$ *is of minimal length. This checking becomes feasible by the reason that one may restrict the study to only relatively few long prefixes and suffixes of* $g(a)$.

## 1 Introduction

Let an integer $k \geq 2$ be fixed. A word is called *k-repetition free*, or in short *k-free*, if it does not contain any non-empty subword of the form $R^k$. Thue [16] showed at the beginning of this century that there exist infinitely long 2-free words over a three letter alphabet . He also showed that 3-free infinite words exist over a binary alphabet. Later, various results concerning avoidability of $k$-repetitions in infinite words have been achieved. In most cases, infinite $k$-free words are constructed by iterating a morphism, which connects closely the theory of repetitions to the theory of L systems; see [14]. Morphisms that preserve the k-freeness of words are sharply characterized in [3,8-11]. A general survey of related results is given in [1]. Fundamental topics are discussed in [12,15].

In [7, p.240] Erdös raised the question whether abelian squares are avoidable in infinitely long words over a four letter alphabet $\Sigma$, i.e., whether there exist infinitely many a-2-free words over $\Sigma$. Later, Pleasants [13] showed that over five letters there exists an infinite word without any abelian square. On the other hand, it is easily seen that over three letters abelian squares cannot be avoided in words of length $\geq 8$. In [6] Entringer et al. showed that every infinite word over a binary alphabet contains arbitrarily long abelian squares. Due to Dekking [4] abelian 4-repetitions [3-repetitions] can be avoided in infinite words over two [three] letters. In recent years the original problem concerning abelian squares over four letters has attracted attention especially in the research of free partially commutative monoids; see for instance [2,5].

The positive solution to the problem posed by Erdös is obtained by showing that the iteration of the endomorphism $g_{85}$, defined in (8) below, yields the desired result. The subscript 85 refers to the length of $g_{85}(a)$ that, indeed, is unexpectedly high to be minimal. The form of the words $g_{85}(x)$, $x \in \Sigma$, i.e., the cyclic permutation of letters in these words; see (2) or (3); was already used by Pleasants [13, p.269] in connection with five letters.

However, $g_{85}(a)$ differs from the one in [13] as regards the relative number of occurrences of letters: here the number of occurrences in $g_{85}(a)$ is different for every letter. In fact, the Parikh vector of $\alpha = g_{85}(a)$, i.e., $(\#_a(\alpha), \#_b(\alpha), \#_c(\alpha), \#_d(\alpha))$, equals $(19, 21, 27, 18)$, whereas the corresponding vector in [13] is $(7, 2, 2, 2, 2)$. It is an open problem whether there exists an a-2-free endomorphism $g: \Sigma^* \to \Sigma^*$, of another form, such that $g(abcd)$ is shorter than $g_{85}(abcd)$. In Lemma 2 we eliminate one possibility that appeared attractive when searching for such a morphism $g$ - or some other method for constructing an infinite a-2-free word.

## 2 Preliminaries

An *alphabet* $\Sigma$ is a finite non-empty set of abstract symbols called *letters*. A *word* over $\Sigma$ is a finite (unless otherwise indicated) string of letters belonging to $\Sigma$, i.e., words [non-empty words] are elements of the free monoid $\Sigma^*$ [the free semigroup $\Sigma^+$] generated by $\Sigma$. The *empty word* is denoted by $\lambda$.

Let $w = x_0 \cdots x_m$, $x_i \in \Sigma$. The *length* of the word $w$, denoted by $|w|$, is the number of occurrences of letters in $w$, i.e., $|w| = m + 1$. The *mirror image* of $w$, denoted by $\mathrm{mir}(w)$, is the word $x_m \cdots x_0$. Let the alphabet considered be $\Sigma = \{a_0,\dots,a_n\}$. The number of occurrences of one letter $x \in \Sigma$ in $w$ is denoted by $\#_x(w)$, or simply by $\#_j(w)$ if $x = a_j$. The notation $\psi(w)$ is used to denote the *Parikh vector* of $w$, i.e., $\psi(w) = (\#_0(w), \dots, \#_n(w))$.

A word $u$ is called a *subword* of a word $w$ if $w = p\,u\,s$ for some words $p$ and $s$. The notation $SW(w)$ stands for the set of all subwords of $w$. If $p = \lambda$ [$s = \lambda$], then $u$ is called a *prefix* [a *suffix*] of $w$. By $PREF(w)$ [$SUFF(w)$] we mean the set of all prefixes [suffixes] of $w$. The notation $pref(w)$ [$suff(w)$] denotes an element in $PREF(w)$ [$SUFF(w)$]. In the case $w \neq \lambda$ we write $first(w)$ [$last(w)$] to denote the first [the last] letter of $w$.

Let $k \geq 2$ be a given integer. A *k-repetition* [an *abelian k-repetition*] is a non-empty word of the form $R^k$ [$P_1 \cdots P_k$, where $\psi(P_\mu) = \psi(P_\nu)$ for all $1 \leq \mu < \nu \leq k$, i.e., $P_i$:s are permutations of each other]. Instead of [abelian] 2- and 3-repetitions terms [*abelian*] *squares* and [*abelian*] *cubes* are often used. A word or an $\omega$-word (explained below) is called *k-repetition free* [*abelian k-repetition free*, or *k-free in the abelian sense*], or in short *k-free* [*a-k-free*], if it does not contain any $k$-repetition [abelian $k$-repetition] as a subword. A word sequence or a set is $k$-free [a-$k$-free] if all words in it are $k$-free [a-$k$-free].

Subsets of $\Sigma^*$ are called *languages* over $\Sigma$. The *catenation* of languages $L$ and $L_1$, denoted by $LL_1$, is the language $\{uv \mid u \in L, v \in L_1\}$. The notation $L^i$, where $L$ is a language and $i \in \mathbb{N}$, is defined as follows: $L^0 = \{\lambda\}$ and $L^{i+1} = L^iL$. Let $L^* = \bigcup_{i=0}^{\infty} L^i$ and $L^+ = \bigcup_{i=1}^{\infty} L^i$. If a language contains only one word, say $w$, then we sometimes write $w$ instead of $\{w\}$; especially, $w^* = \{w^i \mid i \geq 0\}$ and $w^+ = \{w^i \mid i \geq 1\}$.

For a language $L$ we write $Q(L)$, where $Q$ is $SW$, $PREF$ or $SUFF$, to denote the set $\bigcup_{w \in L} Q(w)$. Moreover, for $n$ in $\mathbb{N}$, we write $Q_n(L)$ to denote the words in $Q(L)$ of length $n$. The notation $pref_n(w)$ [$suff_n(w)$] is used for an element in $PREF_n(w)$ [$SUFF_n(w)$].

A *morphism* $h$ is a mapping between free monoids $\Sigma^*$ and $\Sigma'^*$ with $h(uv) = h(u)h(v)$ for every $u$ and $v$ in $\Sigma^*$. Especially, $h(\lambda) = \lambda$. A morphism $h: \Sigma^* \to \Sigma'^*$, being compatible with the catenation of words, is uniquely defined if the word $h(a) \in \Sigma'^*$ is (effectively) given for each $a$ in $\Sigma$. If $\Sigma' = \Sigma$, we call $h$ an *endomorphism* (and usually write $g$ instead of $h$). For a morphism $h$ and a language $L$ we define $h(L) =$

$\{h(w) \mid w \in L\}$. A morphism $h$ is termed *uniformly growing* if $|h(a)| = |h(b)| \geq 2$ for every $a$ and $b$ in $\Sigma$.

For a given integer $k \geq 2$, a morphism $h$: $\Sigma^* \to \Sigma^*$ is called *k-free* [*a-k-free*] if $h(w)$ is $k$-free [a-$k$-free] for every $k$-free [a-$k$-free] $w$ in $\Sigma^*$.

As regards the $L$ systems, we specify the following concepts. A *D0L system* is a triple $G = (\Sigma,g,\alpha_0)$, where $\Sigma$ is an alphabet, $g$: $\Sigma^* \to \Sigma^*$ is an endomorphism and $\alpha_0$, called the *axiom*, is a word over $\Sigma$. The (*word*) *sequence* $S(G)$ generated by $G$ consists of the words

$$\alpha_0 = g^0(\alpha_0), \quad g^1(\alpha_0), \quad g^2(\alpha_0), \quad g^3(\alpha_0), \quad \ldots \quad ,$$

where $g^i(\alpha_0) = g(g^{i-1}(\alpha_0))$ for $i \geq 1$. The *language* of $G$ is defined by $L(G) = \{g^i(\alpha_0) \mid i \geq 0\}$. Languages [sequences] defined by a D0L system are referred to as *D0L languages* [*D0L sequences*]. D0L systems provide a very convenient way for defining languages and infinite words. Furthermore, if $g$ and $\alpha_0$ are $k$-free [a-$k$-free], then the iteration of $g$ will yield a $k$-free [a-$k$-free] D0L sequence.

An *ω-word* is an infinite sequence, from left to right, of letters of an alphabet $\Sigma$. Thus an ω-word can be identified with a mapping of $\mathbb{N}_+$ into $\Sigma$. One can construct an ω-word, for example, by iterating an endomorphism $g$: $\Sigma^* \to \Sigma^*$ such that $\lambda \notin g(\Sigma)$ and $g(a) = aw$ for some $a \in \Sigma$, $w \in \Sigma^+$. Such a morphism $g$ is called *prefix preserving* by the reason that $g^i(a)$ is a proper prefix of $g^{i+1}(a)$ whenever $i \geq 0$. An ω-word is obtained as the "limit" of the sequence $g^i(a)$; $i = 0,1,2,\ldots$ .

In the sequel $\Sigma$ denotes the four letter alphabet $\{a, b, c, d\}$. Especially, for all words $w \in \Sigma^*$ below, the Parikh vector of $w$ will be of the form $\psi(w) = (\#_a(w), \#_b(w), \#_c(w), \#_d(w)) = (\#_0(w), \#_1(w), \#_2(w), \#_3(w))$. In many cases it will be convenient to consider $\Sigma$ as *alphabetically ordered* in the usual way (i.e. $a < b < c < d$) and extend this order to a total order on $\Sigma^+$ (i.e. for any $u$, $v$ in $\Sigma^+$, $u < v$ iff $v$ is in $u\Sigma^+$ or $u = pxs$, $v = pys_1$ with $x < y$; $x, y \in \Sigma$; $p, s, s_1 \in \Sigma^*$. This order on $\Sigma^+$ is called a *lexicographic order* in [12, p.64]).

Let $\sigma$: $\Sigma^* \to \Sigma^*$ be the letter-to-letter endomorphism that performs the cyclic permutation

$$\sigma(a) = b, \quad \sigma(b) = c, \quad \sigma(c) = d \quad \text{and} \quad \sigma(d) = a.$$

The inverse of $\sigma$, denoted by $\sigma^{-1}$, is regarded as a morphism $\sigma^{-1}$: $\Sigma^* \to \Sigma^*$ given by

$$\sigma^{-1}(a) = d, \quad \sigma^{-1}(b) = a, \quad \sigma^{-1}(c) = b \quad \text{and} \quad \sigma^{-1}(d) = c.$$

Thus, informally, assuming the set $\Sigma = \{a,b,c,d\}$ alphabetically ordered, $\sigma$ [$\sigma^{-1}$] gives the immediate cyclic successors [predecessors] of letters in $\Sigma$. We can now use the notation $\sigma^i$ to mean a morphism also in the case where $i$ is a negative integer. Moreover, for every word $w$ in $\Sigma^*$ and integers $i, j$ in $\mathbf{Z}$,

(1) $$\sigma^i(w) = \sigma^j(w) \quad \text{whenever} \quad i \equiv j (\text{mod } 4).$$

Concerning morphisms that generate growing D0L sequences (and ω-words) we consider only uniformly growing endomorphisms $g$: $\Sigma^* \to \Sigma^*$ of the form

(2) $$g(\sigma(x)) = \sigma(g(x)) \quad \text{for all} \quad x \text{ in } \Sigma.$$

Thus, for a given $g(a)$, $g(b) = \sigma(g(a))$, $g(c) = \sigma(g(b))$ and $g(d) = \sigma(g(c))$, that is,

$$(3) \qquad g(b) = \sigma(g(a)), \ g(c) = \sigma^2(g(a)) \text{ and } g(d) = \sigma^3(g(a)).$$

By induction, for a morphism $g$ in (2),

$$g(\sigma^i(x)) = \sigma^i(g(x)) \quad \text{for all} \quad i \text{ in } \mathbf{Z} \text{ and } x \text{ in } \Sigma.$$

From this we see that, whenever $g\colon \Sigma^* \to \Sigma^*$ is of the form (2) and $i$ is in $\mathbf{Z}$, the composite mappings $g\sigma^i$ and $\sigma^i g$ are endomorphisms such that $g\sigma^i = \sigma^i g$. As a summary, recalling (1),

$$(4) \qquad g\sigma^i = \sigma^i g = \sigma^j g \quad \text{for all} \quad i, j \text{ in } \mathbf{Z} \text{ with } i \equiv j (\text{mod } 4).$$

Using (4) we see that the composition of morphisms $\sigma^i$ and $g$ is commutative (and associative). Indeed, it should now be clear that permutation of morphisms in a composition of morphisms $\sigma$ and $g$ gives back the same morphism. For example,

$$(5) \qquad g^{i_0}\sigma^{j_0}g^{i_1}\sigma^{j_1}\cdots g^{i_m}\sigma^{j_n} = g^{i_0+i_1+\cdots+i_m}\sigma^{j_0+j_1+\cdots+j_n} = \sigma^{j_0+j_1+\cdots+j_n}g^{i_0+i_1+\cdots+i_m},$$

whenever exponents $i_\mu$ are in $\mathbf{N}$ and $j_\nu$ in $\mathbf{Z}$.

Let $i$ be an integer in $\mathbf{Z}$. A word $u$ in $\Sigma^+$ is an abelian square iff $\sigma^i(u)$ is an abelian square. Therefore, it is easy to see that, for every word $w$ in $\Sigma^*$,

$$(6) \qquad \sigma^i(w) \text{ is a-2-free} \quad \text{iff} \quad w \text{ is a-2-free.}$$

Let $v$ be an a-2-free word in $\Sigma^*$ and $g$ a morphism of the form (2). By (6), $\sigma^i(g(v))$ is a-2-free iff $g(v)$ is a-2-free. Consequently,

$$(7) \qquad \sigma^i g \ (= g\sigma^i) \text{ is a-2-free} \quad \text{iff} \quad g \text{ is a-2-free.}$$

## 3 Existence

We show that the following prefix preserving endomorphism $g_{85}\colon \Sigma^* \to \Sigma^*$, with $\Sigma = \{a, b, c, d\}$, not only generates an a-2-free $\omega$-word but is also an a-2-free morphism, i.e., $g_{85}(w)$ is a-2-free for every a-2-free $w$ in $\Sigma^*$. The morphism $g_{85}$ being of the same form as $g$ in (2), we have only to give $g_{85}(a)$ in order to define it. A hyphen - indicates concatenation of words.

$$(8) \qquad g_{85}(a) = \begin{array}{l} abcacdcbcdcadcdbdabacabadbabcbdbcbacbcdcacb\text{-} \\ abdabacadcbcdcacdbcbacbcdcacdcbdcdadbdcbca \end{array}$$

Here $|g_{85}(a)| = 85$ and the Parikh vector of $g_{85}(a)$ is $\psi(g_{85}(a)) = (19, 21, 27, 18)$.

**Theorem 1.** *The endomorphism $g_{85}$ defined in (8) is a-2-free.*

**Proof.** Let $u$ in $\Sigma^+$ be a word such that an abelian square is a subword of $g_{85}(u)$. Let $w$ be one of the shortest subwords of $u$ such that $g_{85}(w) = \alpha P_1 P_2 \beta$ for some words

$\alpha$, $\beta$ in $\Sigma^*$ and $P_1$, $P_2$ in $\Sigma^+$ with $\psi(P_1) = \psi(P_2)$. We show that in this situation an abelian square is necessarily a subword of $w$, too. For this purpose we prove that there remains for consideration only finitely many cases, which can be checked with the aid of a computer within a reasonable time. We examined these cases carefully three times using different computing environments: COMMON LISP was used as a programming language on a 68020-based UNIX workstation and on a 80386-based MS-DOS system, and the language C was used on a 80386-based MS-DOS system. Also different methods were used: precalculated lists of Parikh vectors (described below) in connection with LISP, and in connection with C we checked the relevant subwords of every reduced $g_{85}(w)$ (for the term *reduced* see the text below).

We divide in (12) the different possibilities concerning $g_{85}(w) = \alpha\, P_1P_2\, \beta$ above into five main cases. These cases were chosen with intend to make the required computer programs simple. In the sequel, let $x, y, z, t$ be letters in $\Sigma$ and $v_1$, $v_2$ be words in $\Sigma^*$ such that $|v_1| = |v_2|$. Consider the situation $g_{85}(w) = \alpha\, P_1P_2\, \beta$ presented above. Recalling that $w$ in $SW(u)$ was of minimal length it is seen that $g_{85}(w)$ has one of the following structures:

(9) $g_{85}(v_1v_2) = P_1P_2,\quad g_{85}(x\, v_1v_2\, z) = \alpha\, P_1P_2\, \beta$ with $|\alpha| = |\beta|$,

$g_{85}(x\, v_1\, y\, v_2) = \alpha\, P_1P_2,\quad g_{85}(v_1\, y\, v_2\, z) = P_1P_2\, \beta,\quad g_{85}(x\, v_1\, y\, v_2\, z) = \alpha\, P_1P_2\, \beta,$

$g_{85}(x\, t\, v_1\, y\, v_2\, z) = \alpha\, P_1P_2\, \beta$ with $g_{85}(x\, t\, v_1)$ in $PREF(\alpha\, P_1)$ or

$g_{85}(x\, v_1\, y\, v_2\, t\, z) = \alpha\, P_1P_2\, \beta$ with $g_{85}(v_2\, t\, z)$ in $SUFF(P_2\, \beta)$.

Let $w_1$ and $w_2$ be words obtained from words $v_1$ and $v_2$ as follows:

(10) $w_1 = a^{i_0}\, b^{i_1}\, c^{i_2}\, d^{i_3}$ with $i_\mu = \#_\mu(v_1) - \min(\#_\mu(v_1), \#_\mu(v_2))$ and

$w_2 = a^{j_0}\, b^{j_1}\, c^{j_2}\, d^{j_3}$ with $j_\mu = \#_\mu(v_2) - \min(\#_\mu(v_1), \#_\mu(v_2)),\quad \mu = 0,1,2,3.$

Note that here $i_\mu = 0$ or $j_\mu = 0$ for every $\mu = 0,1,2,3$. Informally speaking, $w_1$ and $w_2$ are reduced words of $v_1$ and $v_2$ obtained by firstly rearranging the occurrences of letters in $v_1$ and $v_2$ in alphabetical order and then cancelling, for every letter $x$ in $\Sigma$, occurrences that appear in both words $v_1$ and $v_2$, i.e., $w_1$ and $w_2$ are obtained by erasing the underlined subwords of

$$v_1' = \underline{a^{k_0}}\, a^{i_0}\, \underline{b^{k_1}}\, b^{i_1}\, \underline{c^{k_2}}\, c^{i_2}\, \underline{d^{k_3}}\, d^{i_3} \quad \text{and} \quad v_2' = \underline{a^{k_0}}\, a^{j_0}\, \underline{b^{k_1}}\, b^{j_1}\, \underline{c^{k_2}}\, c^{j_2}\, \underline{d^{k_3}}\, d^{j_3},$$

where $k_\mu + i_\mu = \#_\mu(v_1)$ and $k_\mu + j_\mu = \#_\mu(v_2)$ for every $\mu = 0,1,2,3$. Moreover, there does not exist a letter $x$ in $\Sigma$ occurring in both $w_1$ and $w_2$, since $i_\mu = 0$ or $j_\mu = 0$ in (10). Obviously,

(11) $$|w_1| = |w_2| \quad \text{and} \quad \psi(w_1) - \psi(w_2) = \psi(v_1) - \psi(v_2).$$

Any words $w'$ and $w''$ in $\Sigma^*$ are referred to as *reduced*, and the pair $(w', w'')$ as a *reduced pair*, if there does not exist any letter $x$ in $\Sigma$ that occurres in both $w'$ and $w''$.

Below we consider the reduced cases of $g_{85}(w)$ in (9). Instead of the abelian square $P_1P_2$ we now write $R_1R_2$. Illustration of these cases is as follows:

(12)

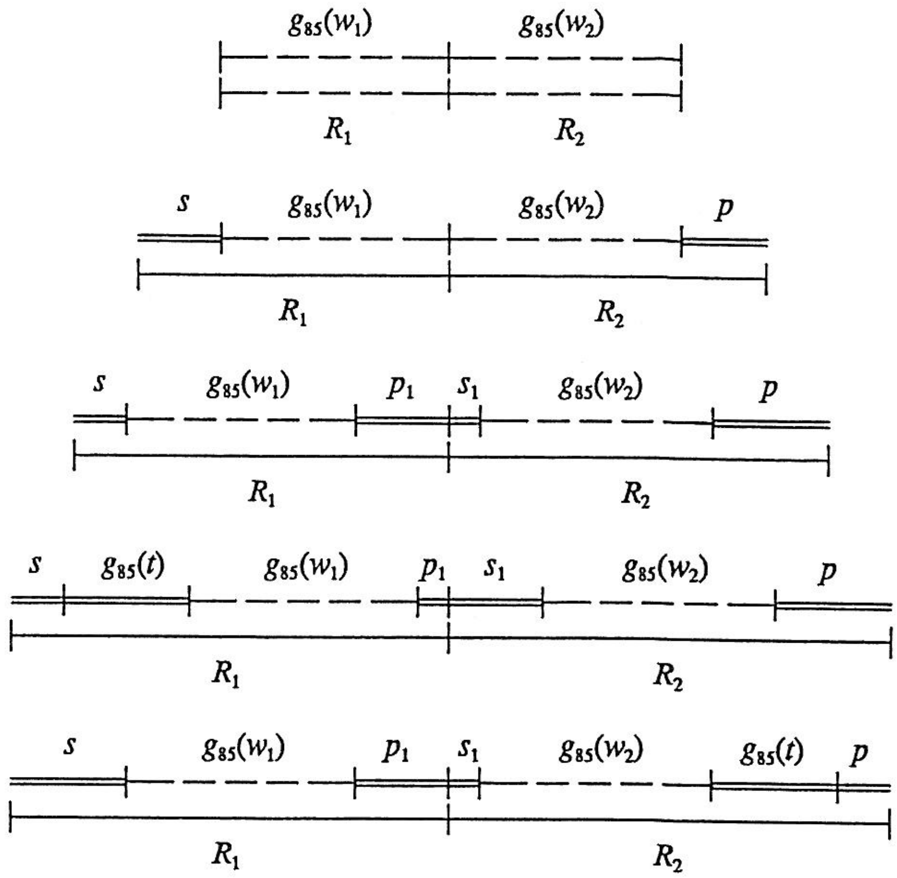
(i)

(ii)

(iii)

(iv)

(v)




where in accordance with (9) $\alpha s = g_{85}(x)$, $p\beta = g_{85}(z)$, $p_1 \neq \lambda$, $s_1 \neq \lambda$. Moreover, in greater detail: in (12, ii) $0 < |s| = |p| < |g_{85}(a)|$; in (12, iii) $0 \leq |s| \leq |g_{85}(a)|$, $0 \leq |p| \leq |g_{85}(a)|$; in (12, iv) $0 < |s| < |g_{85}(a)|$, $0 < |p| \leq |g_{85}(a)|$; and in (12, v) $0 < |s| \leq |g_{85}(a)|$, $0 < |p| < |g_{85}(a)|$.

Let $\psi_D(w', w'')$ denote the difference vector $\psi(g_{85}(w')) - \psi(g_{85}(w''))$, where $w'$ and $w''$ are any words in $\Sigma^*$. In the cases (ii) - (v) of (12) the existence of an abelian square $R_1R_2$ implies that the difference vectors

(13) $D_{\text{ii}} = \psi(p) - \psi(s)$, $D_{\text{iii}} = \psi(s_1 p) - \psi(s p_1)$, $D_{\text{iv}} = \psi(s_1 p) - \psi(s\, g_{85}(t)\, p_1)$ and

$D_{\text{v}} = \psi(s_1\, g_{85}(t)\, p) - \psi(s\, p_1)$

are equal to $\psi_D(w_1, w_2)$, where $w_1$ and $w_2$ are the reduced words corresponding each case (ii) - (v) in question. Note, for example, that in (12.ii) $\psi(R_1) = \psi(R_2)$ means that $\psi(s\, g_{85}(w_1)) = \psi(g_{85}(w_2)p)$, i.e., $\psi(s) + \psi(g_{85}(w_1)) = \psi(g_{85}(w_2)) + \psi(p)$ giving $D_{\text{ii}} = \psi(p) - \psi(s) = \psi(g_{85}(w_1)) - \psi(g_{85}(w_2)) = \psi_D(w_1, w_2)$. Using precalculated lists of Parikh vectors for prefixes (*preflist.x*) and suffixes (*sufflist.x*) of each $g_{85}(x)$, with $x$ in $\Sigma$, one may quickly check with the aid of a computer that

(14) $m_{\text{ii}} = 9$, $m_{\text{iii}} = 15$, $m_{\text{iv}} = 18$ and $m_{\text{v}} = 19$,

where $m_\mu = \max\{|D_{\mu,k}| \mid D_\mu$ goes through all the values in the case $\mu$ (all combinations for $s, p_1, s_1, p$ and $t$ are included - but neither $xt$ nor $tz$ is a square) and $k = 0,1,2,3\}$

($D_{\mu,k}$ is the $k+1$ th component of $D_\mu = (D_{\mu,0}, D_{\mu,1}, D_{\mu,2}, D_{\mu,3})$). In order to explain the precalculated lists *preflist.x* and *sufflist.x* mentioned above we give two examples:

$$preflist.a \quad = \quad ((1,0,0,0)\ (1,1,0,0)\ (1,1,1,0)\ \ldots\ (19,21,27,18))$$

$$sufflist.a \quad = \quad ((19,21,27,18)\ (18,21,27,18)\ (18,20,27,18)\ \ldots\ (1,0,0,0))$$

The reader may compare these lists with $g_{85}(a)$ in (8).

Considering the absolute values of components of the difference vectors $\psi_D(w_1, w_2)$, with $w_1$, $w_2$ as above, let $m(w', w'') = \max\{|(\psi_D(w', w''))_k| \mid k = 0,1,2,3\}$, where $w'$ and $w''$ are any words in $\Sigma^*$. Straightforward calculation (ignoring here firstly (12) and (14)) shows that concerning all possible reduced pairs $(w', w'')$ in the case $1 \leq |w'| = |w''| \leq 4$, $m(w', w'') \leq 19$ the difference vector $\psi_D(w', w'')$ can have only the values:

| | | | | | |
|---|---|---|---|---|---|
| (15) $\psi_D(a, b)$ | = | (1, 2, 6, -9) | $\psi_D(ab, cd)$ | = | (-11, -5, 11, 5) |
| $\psi_D(a, c)$ | = | (-8, 3, 8, -3) | $\psi_D(ac, bd)$ | = | (7, -7, 7, -7) |
| $\psi_D(aa, bb)$ | = | (2, 4, 12, -18) | $\psi_D(aab, ccd)$ | = | (-19, -2, 19, 2) |
| $\psi_D(aa, bc)$ | = | (-7, 5, 14, -12) | $\psi_D(aac, bbd)$ | = | (8, -5, 13, -16) |
| $\psi_D(aa, bd)$ | = | (-1, -4, 15, -10) | $\psi_D(aad, bcc)$ | = | (-13, 14, 13, -14) |
| $\psi_D(aa, cc)$ | = | (-16, 6, 16, -6) | $\psi_D(aacc, bbdd)$ | = | (14, -14, 14, -14) |
| $\psi_D(aa, cd)$ | = | (-10, -3, 17, -4) | | | |

and, in addition, the values obtained by cyclic permutation of the vectors $\psi_D(w', w'')$ in (15) and their opposite vectors $-\psi_D(w', w'')$, i.e., in the case $1 \leq |w'| = |w''| \leq 4$, $m(w', w'') \leq 19$, with reduced $w'$ and $w''$, the difference vector $\psi_D(w', w'')$ can have at most 72 (as one can check) different values that are among the vectors $\pm\psi_D(\sigma^i(w'), \sigma^i(w''))$; $i = 0,1,2,3$; where $(w', w'')$ goes through the reduced pairs expressed by (15). In the end of the proof it will be shown that for every reduced pair $(w', w'')$, with $|w'| = |w''| \geq 5$, one has $m(w', w'') \geq 21$, implying, by (13) and (14), that the case $|w_1| = |w_2| \geq 5$ is contradictory in (12).

Let $|w_1| = |w_2| \leq 4$ in (12). Taking (13), (14) and (15) into account one may now, with the aid of a computer, list all the cases that might exist in (12) (note that also the possibility $w_1 = w_2 = \lambda$ is relevant in the cases (ii) - (v) of (12)). The list of the cases is rather long, but, on the other hand, it is straightforward to check that it contains only those cases in which all of the structures of $g_{85}(w) = \alpha P_1P_2\beta$, presented in (9), obtained by rearranging the letters of $w_1$ and $w_2$, or by rearranging the letters and adding some letters to $w_1$ and $w_2$ (the same letters to both), are such that $w$ contains an abelian square. In other words, in every case, the pair $(w_1, w_2)$ is not a reduced pair of any pair $(v_1, v_2)$ such that $g_{85}(w) = \alpha P_1P_2\beta$ has one of the structures of (9) and $w$ is a-2-free. This means that if $|w_1| = |w_2| \leq 4$, then $g_{85}(w)$ is a-2-free whenever $w$ is a-2-free. We give two detailed examples of the cases concerning (iii) and (iv) in (12):

$$(16) \qquad g_{85}(a\, w_1\, a\, w_2\, b) \;=\; \alpha\, R_1R_2\, \beta;$$

with $w_1 = b$, $w_2 = a$, $|\alpha| = 72$, $|\beta| = 1$, $|R_1| = |R_2| = 176$; and

$$(17) \qquad g_{85}(a\, b\, w_1\, c\, w_2\, d) \;=\; \alpha\, R_1R_2\, \beta;$$

with $w_1 = cd$, $w_2 = ab$, $|\alpha| = 13$, $|\beta| = 5$, $|R_1| = |R_2| = 331$. The reader may easily get convinced that in (16) and (17) the replacement of the subwords $w_1$ and $w_2$ of $a\,w_1\,a\,w_2\,b$ and $a\,b\,w_1\,c\,w_2\,d$ by any $v_1$ and $v_2$ such that $(w_1, w_2)$ is a reduced pair of $(v_1, v_2)$ gives only words that are not a-2-free. It turns out that the computer program will list only those cases in which $|w_1| = |w_2| \leq 2$. This makes the cases very straightforward to check (we assume that the reader does not want to leave the whole work to the computer). Moreover, it is possible to reduce the case in (17) to a shorter one by considering the reduced pair $(cd, a)$ instead of the pair $(b\,w_1, w_2) = (b\,cd, ab)$. Permitting this kind of reduction leaves only the cases $|w_1| = |w_2| \leq 1$ to check!

Let $|w_1| = |w_2| \geq 5$ in (12). We show that in this case always $m(w_1, w_2) \geq 21$. This contradiction with (13) and (14) finally makes the proof complete. We write in short $\Delta$ to denote $\psi_D(w_1, w_2)$. Without loss of generality we may assume that

$$(18) \qquad w_1 = a^{i_0}\, b^{i_1}\, c^{i_2}\, d^{i_3} \quad \text{with} \quad i_0 = \max\{i_0, i_1, i_2, i_3\}.$$

Indeed, the other cases concerning $\max\{i_0, i_1, i_2, i_3\}$ are readily obtained from the case in (18) by considering the words $\sigma^k(w_1)$ and $\sigma^k(w_2)$; with $k = 1,2,3$; instead of $w_1$ and $w_2$, since $m(\sigma^k(w_1), \sigma^k(w_2)) = m(w_1, w_2)$. Firstly, recall that

$$(19) \qquad \begin{aligned} \psi(g_{85}(a)) &= (19, 21, 27, 18) \\ \psi(g_{85}(b)) &= (18, 19, 21, 27) \\ \psi(g_{85}(c)) &= (27, 18, 19, 21) \\ \psi(g_{85}(d)) &= (21, 27, 18, 19), \end{aligned}$$

where the shading of the columns is used to indicate that we use these columns in order to get a lower bound for $m(w_1,w_2)$. Note also that $|\Delta_k|$; with $k = 0,1,2$ or $3$; refers to the difference concerning the occurrences of a letter $\sigma^k(a)$, i.e., it refers to the $k+1$ th shaded column of (19). If $i_0 = 2$ in (18), then $w_1$ contains three different letters and, consequently, $w_2 = (\sigma^{\mu}(a))^j$, for some $\mu \in \{1,2,3\}$ and $j = |w_1| \geq 5$, implying that $|\Delta_\nu| > j\cdot|21 - 27| \geq 30$, where $\nu \equiv (2 + \mu) \pmod 4$ and $\nu \in \{0,1,2,3\}$. A similiar situation is obtained if $i_0 = |w_1|$. Thus we may assume that in (18) one has $3 \leq i_0 < |w_1|$, i.e.,

$$(20) \qquad w_1 \in a^{i_0}\,(b^+ \cup c^+ \cup d^+) \quad \text{with} \quad i_0 = \max\{i_0, i_1, i_2, i_3\} \geq 3.$$

Consider separately the three cases of (20).

Let $w_1 = a^{i_0}\, b^{i_1}$. Then $w_2 = c^{j_2}\, d^{j_3}$ with $j_2 + j_3 \geq 5$. Thus $|\Delta_2| = |i_0\cdot 27 + i_1\cdot 21 - j_2\cdot 19 - j_3\cdot 18| > i_0\cdot(27 - 19) + i_1\cdot(21 - 19) \geq 3\cdot(27 - 19) + 2\cdot(21 - 19) = 28$.

Let $w_1 = a^{i_0}\, c^{i_2}$. Then $w_2 = b^{j_1}\, d^{j_3}$ with $j_1 + j_3 \geq 5$. If $j_1 < j_3$, then $|\Delta_2| = |i_0\cdot 27 + i_2\cdot 19 - j_1\cdot 21 - j_3\cdot 18| \geq 3\cdot(27 - 18) + 2\cdot(19 - 21) = 23$. On the other hand, if $j_1 \geq j_3$, then $|\Delta_3| = |i_0\cdot 18 + i_2\cdot 21 - j_1\cdot 27 - j_3\cdot 19| \geq |3\cdot(18 - 27) + 3\cdot(21 - 19)| = 21$.

Let $w_1 = a^{i_0}\, d^{i_3}$. Then $w_2 = b^{j_1}\, c^{j_2}$ with $j_1 + j_2 \geq 5$. If $j_1 = 1$, then $j_2 \geq 4$ and $|\Delta_0| = |i_0\cdot 19 + i_3\cdot 21 - j_1\cdot 18 - j_2\cdot 27| \geq |3\cdot(19 - 27) + (21 - 27) + (21 - 18)| = 27$. On the other hand, if $j_1 \geq 2$, then $|\Delta_3| = |i_0\cdot 18 + i_3\cdot 19 - j_1\cdot 27 - j_2\cdot 21| \geq |2\cdot(18 - 27) + (18 - 21) + 2\cdot(19 - 21)| = 25$.

Thus we see that if $|w_1| = |w_2| \geq 5$ in (12), then always $m(w_1, w_2) \geq 21$, since $m(w_1, w_2) \geq |\Delta_k|$ for every $k = 0,1,2,3$. Recalling (13), (14) and the result concerning the case $|w_1| = |w_2| \leq 4$, we obtain that $g_{85}(w)$ is of the form $\alpha\, P_1 P_2\, \beta$ only if $w$ contains an abelian square. This proves Theorem 1. □

# 4 Minimality of $|g(a)|$

We study only the case $\Sigma = \{a, b, c, d\}$ and endomorphisms $g: \Sigma^* \to \Sigma^*$ of the form (2). In Lemma 5 below it is stated that, for a fixed length of $g(a)$, there exists an a-2-free DOL sequence generated by $g$ only if that is the case for $g$ with

$$g(a) = a\,b\,c\,w\,a \quad \text{for some } w \text{ in } \Sigma^+. \tag{21}$$

Consequently, for every fixed length of $g(a)$, when searching candidates for a-2-free endomorphisms (of the form (2)) over $\Sigma$ or a-2-free DOL sequences generated by $g$, it will be sufficient to consider only the morphisms $g$ of the form (21), i.e., only those words $g(a)$ which begin with $abc$ and end with $a$. The proof of Lemma 5 is based on four auxiliary results presented below. Due to space limitation only some outlines for the proofs are given. After Lemma 5, we discuss in detail the method with the help of which the morphism $g_{85}$ in (8) was found.

**Lemma 1.** *Let $g_a$ and $g$ be endomorphisms of the form* (2) *and let $g_a(a) = g(x)$ for the letter $x$ such that $first(g(x)) = a$. Then*
(i) *$g_a$ is a-2-free iff $g$ is a-2-free;*
(ii) *the DOL sequence $S(G_a)$ generated by $G = \{\Sigma, g_a, \sigma^k(\alpha_0)\}$ is a-2-free for every $k$ in $\{0,1,2,3\}$ iff the DOL sequence $S(G)$ generated by $G = \{\Sigma, g, \alpha_0\}$ is a-2-free.*

The following lemma tells us that it is not possible to construct an a-2-free $\omega$-word $\gamma$ over $\Sigma$ in such an attractive way that after starting with a letter, say $a$, every second letter of $\gamma$ would be in a certain subset of $\Sigma$, say in $\{a,c\}$, and the remaining letters in between would be in the complement set $\{b,d\}$, i.e., every prefix of $\gamma$ of even length would be in $(\{a,c\}\,\{b,d\})^*$. For example, the following effort to start such a $\gamma$ will fail:

$$a\,b\,c\,d\,c\,b\,a\,b\,c\,b\,a\,d\,a\,b\,a\,d\,c\,d\,a\,b\,a\,d\,a \ldots\ .$$

**Lemma 2.** *Let $\Sigma_e$ and $\Sigma_o$ be disjoint two letter alphabets such that $\Sigma_e \cup \Sigma_o = \Sigma = \{a,b,c,d\}$. There does not exist any a-2-free $\omega$-word $\gamma$ over $\Sigma$ such that*

$$\gamma = x_0 x_1 x_2 \cdots \quad \textit{with} \quad x_{2i} \in \Sigma_e \quad \textit{and} \quad x_{2i+1} \in \Sigma_o \quad \textit{for every} \quad i \in \mathbb{N}. \tag{22}$$

One may prove Lemma 2 by studying different possibilities for the subwords of $\gamma$. In this approach no computer is needed. Another proof is obtained by using a computer. One finds out that there exist only finitely many words that could be prefixes of an a-2-free $\gamma$ of the form (22). Indeed, it turns out that a word in $\{ac, bd, ca, db\}$, i.e., a word of the form $x\,\sigma^2(x)$, $x \in \Sigma$, is a subword of every a-2-free word $w \in \Sigma^+$ of length 42. This way of proving Lemma 2 was found by Matti K. Sinisalo, University of Oulu. Moreover, he found out that a word in $\{ab, bc, cd, da\}$, i.e., a word of the form $x\,\sigma(x)$, $x \in \Sigma$, is a subword of every a-2-free word $w \in \Sigma^+$ of length 45.

By utilizing Lemma 1 and Lemma 2 (or the results of Sinisalo) one may prove

**Lemma 3.** *Let $g$ be an endomorphism of the form* (2) *and assume that the DOL sequence $S(G)$ generated by $G = \{\Sigma, g, \alpha_0\}$ is a-2-free. Then $first(g(a)) = last(g(a))$.*

Informally, the proof of Lemma 3 is obtained by considering the subwords $w$ of length 2 ocurring in the words of $S(G)$ and by studying the image words $g(w)$ and $g^2(w)$. The same holds for the proof of the following

**Lemma 4.** *Let $g$ be a given endomorphism of the form (2) such that, for some $\alpha_0$ in $\Sigma^+$, there exists an a-2-free DOL sequence $S(G)$ generated by $G = \{\Sigma,g,\alpha_0\}$. Then*

(23) $$SW_2(g(\{a,b,c,d\})) = \bigcup_{i=0}^{3} \sigma^i(\{ab,ac,ad\}).$$

By using a computer one may check that there does not exist any candidate for $g$ if $|g(a)| < 10$. Similarly, it is seen that the set $SUFF_{10}(g(a))\, PREF_{10}(g(\{b,c,d\}))$ is a-2-free only if $g(a)$ is of the form $a\,b\,c\,w\,a$, $a\,d\,c\,w\,a$, $a\,w\,c\,b\,a$ or $a\,w\,c\,d\,a$ for some $w$ in $\Sigma^+$. These facts and the previous lemmas enable us to prove

**Lemma 5.** *Let $g$ and $g_1$ be endomorphisms of the form (2) such that $|g(a)| = |g_1(a)|$ and*

(24) $$g_1(a) = a\,b\,c\,w\,a \quad \textit{for some } w \textit{ in } \Sigma^+.$$

*Then there exists an a-2-free DOL sequence obtained by the iteration of $g$ only if the same holds true for $g_1$, too.*

In what follows we explain how the a-2-free morphism $g_{85}$ in (8) was found and why the image word $g_{85}(a)$ is of minimal length as regards not only a-2-free morphisms (of the form (2)) but also morphisms (of the considered form) that generate an a-2-free DOL sequences.

By the result of Lemma 5, we now see that when searching an endomorphism $g$ of the form (2) that generates an a-2-free DOL sequence, one does not lose any possibilities, if one assumes that $g(a)$ is of the form $a\,b\,c\,w\,a$. Let this be the form of $g(a)$ below. Now one may check by a computer that there does not exist any morphism $g$ of the form (2) such that $|g(a)| \leq 20$ and the set $g(\{ab,ac,ad\})$ is a-2-free. This can be done, e.g. letting, for every fixed length of $g(a) \leq 20$, the word $w \in \Sigma^+$ run, in alphabetical order, through all a-2-free combinations in the word $g(a) = a\,b\,c\,w\,a$ and testing, for every $w$, whether the words $g(a)\, pref_3(g(b)) = a\,b\,c\,w\,a\,b\,c\,d$, $g(ab)$, $g(ac)$ and $g(ad)$ are a-2-free. Using $g(a)\, pref_3(g(b))$ instead of $g(a)$ or $g(ab)$ alone helps in finding an abelian square before the construction of the word $g(ab) = g(a)\,b\,c\,d\,\sigma(w)\,b$, where $\sigma(w)$, or, in fact, some suffix of $\sigma(w)$, has to be computed for every new $w$. After this, one can compute that there exist only 413 words $u = ps$ such that $|p| = |s| = 20$ and $g(a)$ is of the form $p\,w\,s$; with $abc$ in $PREF(p)$ and $last(s) = a$; and the set

$$SUFF_{20}(g(a))\, PREF_{20}(g(\{b,c,d\}))$$

is a-2-free. Below we denote by $p_i$ and $s_j$ the words such that $g(a)$ is of the form $p_i\, w\, s_j = abc\, w'\, a$ and $|p_i| = i$, $|s_j| = j$. Next, from these 413 words $p_{20}s_{20}$, one sees that there remain for consideration only 9 words $p_{10}s_{10}$; recall that the case $|g(a)| \leq 20$ was already checked. By using these 9 words $p_{10}s_{10}$, one may check that there does not exist any morphism $g$ such that $21 \leq |g(a)| \leq 40$ and the set $g(\{ab,ac,ad\})$ is a-2-free. Here too, in order to save computer time, the program should firstly fix the length of $g(a)$, then choose the word $p_{10}s_{10}$ and after that, for every proper canditate $w$, firstly check the a-2-freeness of $g(a)\ pref_{10}(g(b)) = p_{10}\ w\ s_{10}\ \sigma(p_{10})$ (we tried also some other methods but in general without a success in making the computation notably faster). Next, one can reduce the number of the words $p_{20}s_{20}$ in the list that above contained 413 of them. Studying the list of words $p_{30}s_{30}$ such that the words $s_{30}\ \sigma^i(p_{30})$; $i = 1,2,3$; are a-2-free, i.e., the set $SUFF_{30}(g(a))\ PREF_{30}(g(\{b,c,d\}))$ is a-2-free, (the list is too large to be useful as such), one finds that no more than 95 words $p_{20}s_{20}$ are possible for the desired $g(a) = p_{20}\ w\ s_{20}$. By using these 95 words $p_{20}s_{20}$, one may now check that there does not exist any morphism $g$ such that $41 \le |g(a)| \le 60$ and the set $g(\{ab,ac,ad\})$ is a-2-free. Moreover, one may, by using the reduced list of 95 words $p_{20}s_{20}$, produce the list of possible words $p_{23}s_{23}$. After many reduction experiments this list was found to contain no more than 367 words. We also found that the length 46 of the words $p_i s_j$ was nearly optimal for future calculations (at least in our computing environments). This seems to be due to the fact that there remain too many possibilities for longer words $p_i s_i$; with $i > 23$. Therefore, when using these long words $p_i s_i$, one may not get any proper speed-up for the search of promising $g$. One could perhaps get some advantage by using words $p_i s_j$; with $i < j$; since there seems to be more different prefixes than suffixes in the lists containing the words $p_i s_i$. For instance, by studying the prefixes and suffixes in the non-reduced list of words $p_{30}s_{30}$; with a-2-free $s_{30}\ \sigma^i(p_{30})$ for $i = 1,2,3$; one finds that there remain only 10 words $p_{10}s_{20}$ giving for $g(a) = p_{10}\ w\ s_{20}$ the following possibilities:

$$\begin{aligned} g(a) &= abcacdc\ u_1\ \ w\ u_2\ u_3, \\ g(a) &= abcacdcbcd\ w\ bcdcacdcbdcdadbdcbca, \quad \text{or} \\ g(a) &= abcbdadcbc\ w\ u_4\ acdadbdcdbdabdbca; \end{aligned}$$

with $u_1$ in $\{acb, bca, bcd\}$, $u_2$ in $\{abc, bac\}$, $u_3 = babdbcdbdadbdcbca$, $u_4$ in $\{bad, cbc, dbc\}$ and $w$ in $\Sigma^+$. Note here that our choice $abc$ as the prefix of $g(a)$ creates an asymmetric situation because of the permutation $\sigma$ in (2).

Next, one may check all the cases for morphisms $g$ with $61 \le |g(a)| \le 85$. Here the computer time seems to grow proportionally to $m^k$; with $1.4 < m < 1.5$ and $k = |g(a)|$. We found out that not until in the case $|g(a)| = 64$ the set $g(\{ab,ac,ad\})$ is a-2-free, i.e., the set $\bigcup_{i=0}^{3} \sigma^i(\{ab,ac,ad\})$ is a-2-free. Going further, let us say that $g$ passes *Test* $n$, if $g(u)$ is a-2-free for every a-2-free $u$ in $\Sigma^+$ of length $\le n$. It turns out that not until in the case $|g(a)| = k$ there exists $g$ passing *Test* $n$, where $k$ and $n$ form pairs $(k, n)$ as follows: (64, 2), (71, 3), (83, 4) and (85, 5). Moreover, all these morphisms $g$, except $g_{85}$ considered in the previous chapter (the case (85, 5)), are such that they do not generate any a-2-free D0L sequence; this is quite easy to check by Lemma 4 and by the reason that not very many morphisms $g$ pass *Test* 3 (we found about one hundred of them in the case $71 \le |g(a)| \le 100$; only part of the cases were tested for $|g(a)| > 85$). Including $g_{85}$, only two morphisms $g$ passed *Test* 4! The first one gives the pair (83,4) above and is as follows:

$$\begin{aligned} g_{83}(a) \quad = \quad & abcacdcbcdcadbdcadabaca\underline{badbab}cbdbcbacbcdca- \\ & cbacadabdbcdcacdbcbacbcdcacdcbdcdadbdcbca \end{aligned}$$

This morphism does not pass *Test* 5, since, e.g. $g_{83}(acbad)$ is not a-2-free. Moreover, for the underlined subword $u = badbab$ of $g_{83}(a)$, $g_{83}(u)$ is not a-2-free. Consequently, $g_{83}$ does not generate any a-2-free D0L sequence either. The Parikh vector for $g_{83}(a)$ is $\psi(g_{83}(a)) = (19, 20, 26, 18)$.

We carried out the search for $g$ in the case $|g(a)| \le 73$ using two programming languages LISP and C, which ran in parallel and executed the same tests twice on one 68020-based UNIX workstation and on 80286- and 80386-based MS-DOS systems. In the case $74 \leq |g(a)| \leq 85$ only MS-DOS (mainly 80286-based) systems and C language were used. These computer tests were performed in the summer 1990 at the Rovaniemi Institute of Technology. It is assumed that the same tests could be now (spring 1992) repeated by using, say 10, modern microcomputers in a few days. In addition, we let the computers check, for certain prefixes and suffixes of $g(a)$, the cases $86 \leq |g(a)| \leq 100$. In spite of the fact that the time for this checking was as long as in the case $|g(a)| \leq 85$, we did not find any more morphisms $g$ that would pass *Test* 4. Consequently, $g_{85}$ presented in (8) is of rare type. Excluding renaming letters, mirror images of $g_{85}(a)$ and powers of $g_{85}$, we do not know any other endomorphism over a four letter alphabet that is a-2-free or generates an a-2-free D0L sequence.

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